



Good

Evening

Everyone 😊

2D MATRICES

Java

```
int[][] mat = new int[N][M];
```

Python:

```
mat = [[0] * M for _ in range(N)]
```

mat

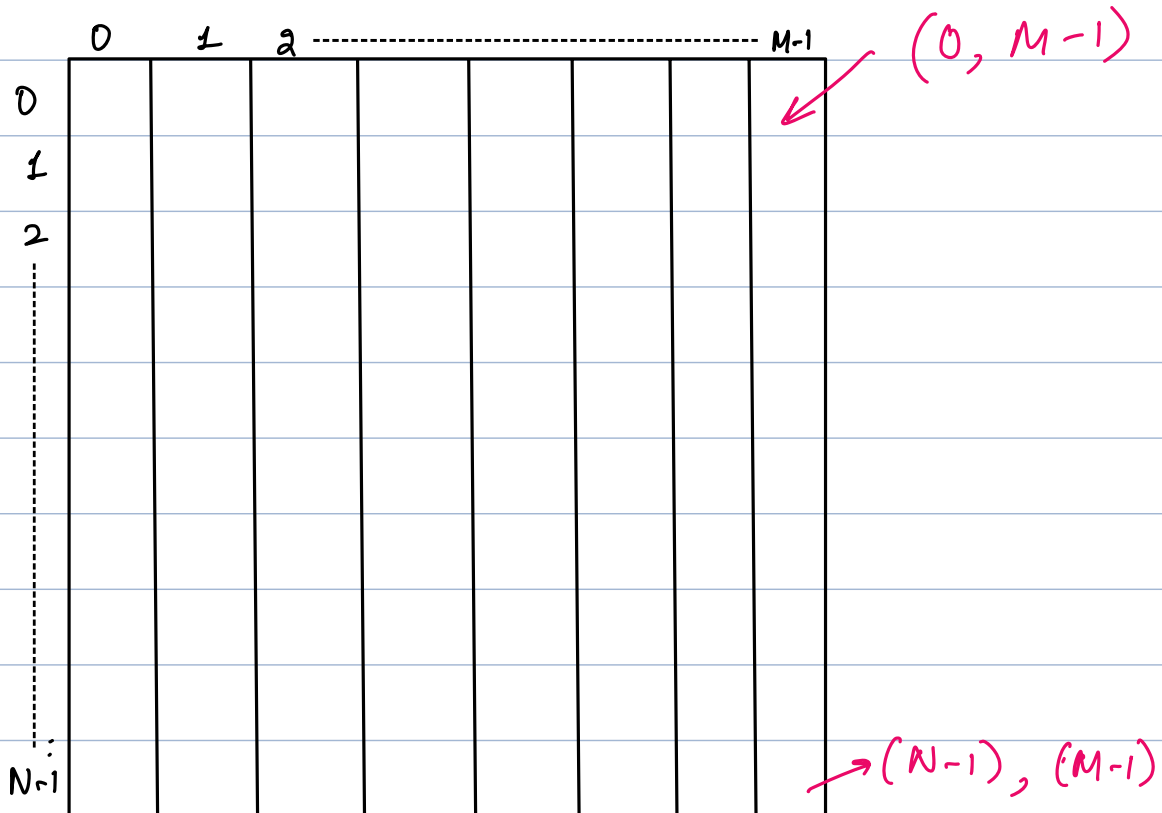
	0	1	2
0			(0,2)
1		(1,1)	
2			

↑
Row
Ind.

Column index

N, M

N x M element



①

Given a 2D matrix $mat[N][M]$, print row wise sum

Testcase

	0	1	2	3	
0	1	2	3	4	⇒
1	5	6	7	8	
2	9	10	11	12	

0	1	2
10	26	42

function SumRow (int mat[N][M])

TC: $O(NM)$

SC: $O(1)$

for ($i \rightarrow 0$ to $N-1$) {

Sum = 0

for ($j \rightarrow 0$ to $M-1$) {

Sum = Sum + mat[i][j]

}

print (sum);

}

}

2

Given a 2D matrix $mat[N][M]$, print col-wise sum

$mat[][] \Rightarrow$

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

15 18 21 24

function SumColumn (int mat $[N][M]$) {

TC: $O(NM)$
SC: $O(1)$

for ($j \rightarrow 0$ to $M-1$) {

Sum = 0

for ($i \rightarrow 0$ to $N-1$) {

Sum = Sum + $mat[i][j]$

}
print (Sum)

}

}

③

Given a matrix as square matrix $mat[N][N]$,
print diagonals.

1. Principal Diagonal :- from top left to bottom right

1	5	8
4	3	1
6	5	2

Anti - Diagonal → from top right to bottom left

1	5	8
4	3	1
6	5	2

lets print principal diagonal :-

	0	1	2
0	1	2	3
1	5	6	7
2	9	10	11

1, 6, 11

TC : $O(N)$

code :-

function printDiagonal (mat[N][N]) {

 i = 0

 while (i < N) {

 print (mat[i][i])

 i = i + 1

 }

}

Anti-Diagonal

	0	1	2
0	1	2	3 ₀₂
1	5	6 ₍₁₁₎	7
2	9 ₍₂₀₎	10	11

3, 6, 9.

$$i = 0$$

$$j = N - 1$$

$$TC : O(N)$$

$$SC : O(1)$$

function Print Anti Diagonal (mat[N][N]) {

$$i = 0, j = N - 1$$

// to print right

while (i < N) {

 print (mat[i][j])

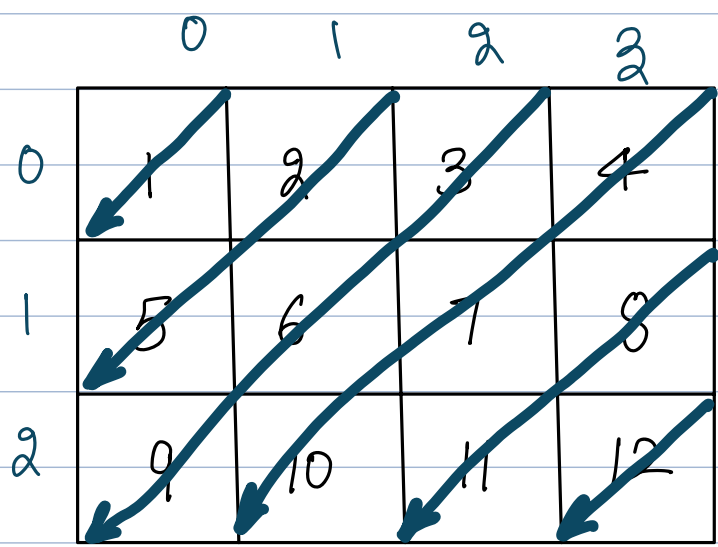
 i++ ;

 j-- ;

}

}

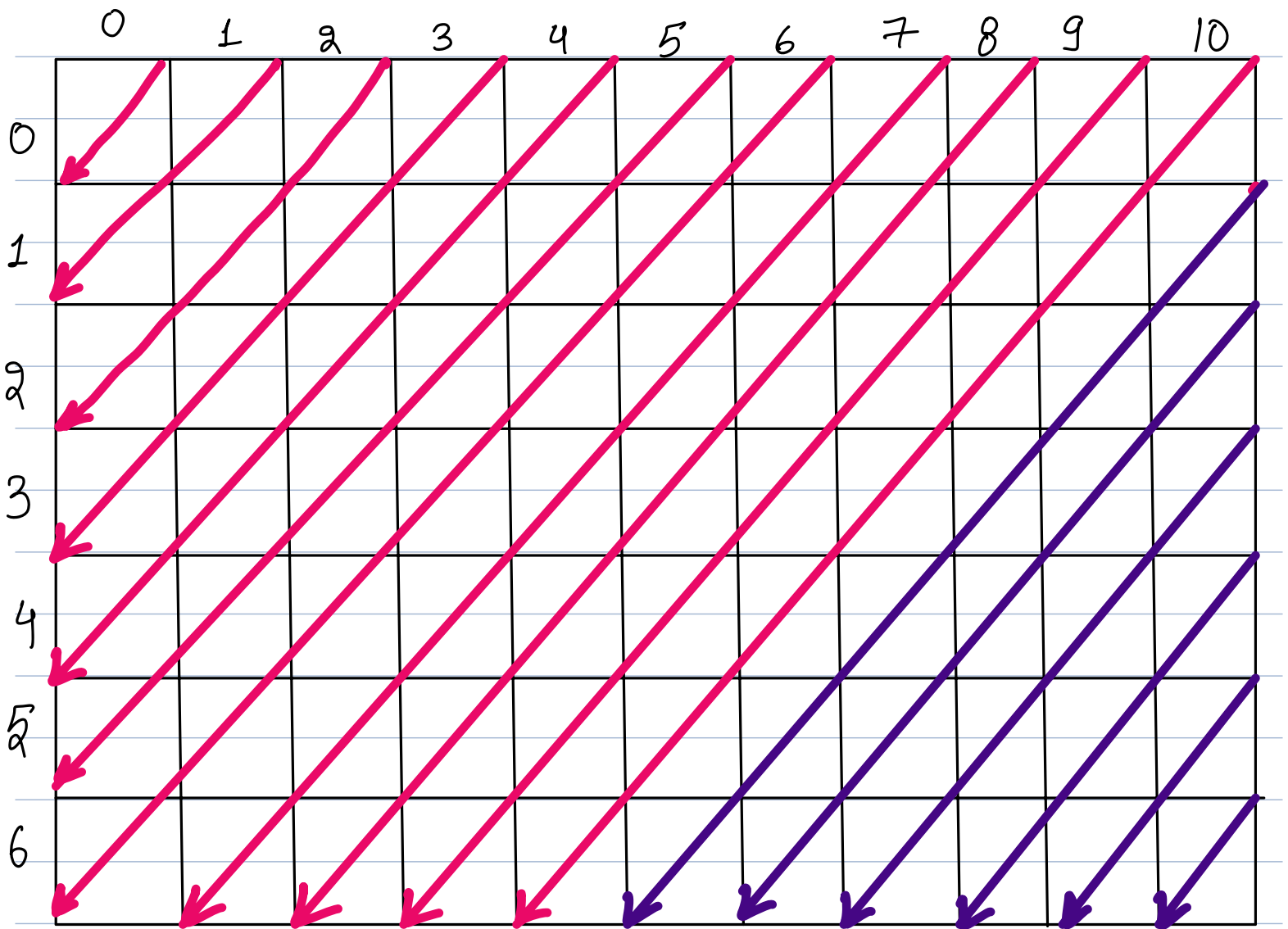
④ Print diagonals in rectangular matrix (right to left)



3x4

1
2 5
3 6 9
4 7 10
8 11
12

⇒ ⑥ diagonals



$$M + N - 1$$

$$7 \times 11$$

$$\text{Total} = M + N - 1$$

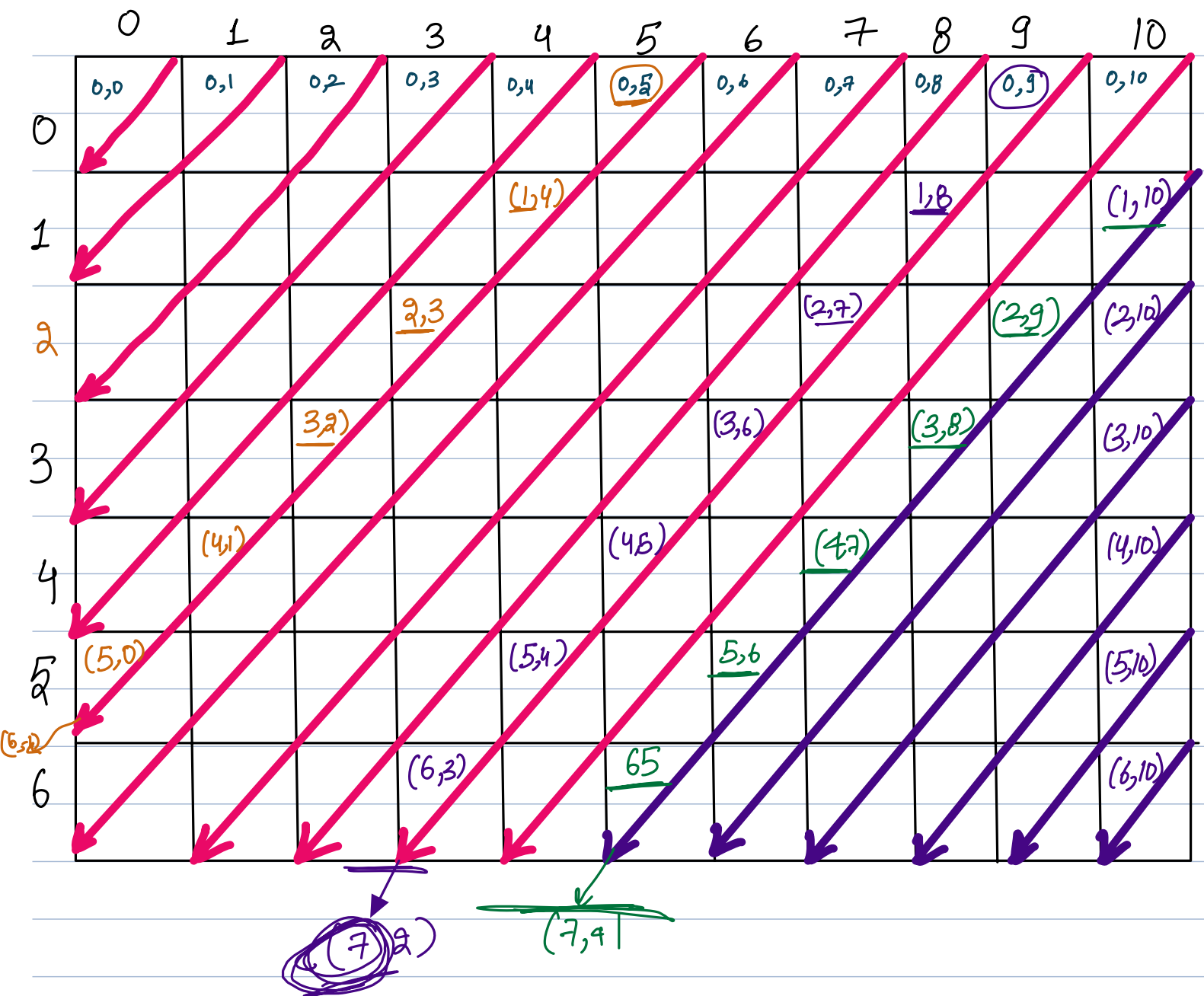
$$7 + 11 - 1 = 17$$

Q: 4×5 ?

$\Rightarrow$

$$4 + 5 - 1$$

$$\Rightarrow 8.$$



Code:

TC: $O(N \times M)$

SC: $O(1)$

function print-diagonals (A[M][N])

// print all diagonals starting at row 0.

i = 0

for (j $\Rightarrow$ 0 to M-1) {

 s = i

 e = j

 while (s < N && e >= 0) {

 print (A[s][e])

 s++

 e--

 }

 print (newline)

}

// print all diagonals start at last column

j = M-1

for (i $\rightarrow$ 1 to N-1) {

 s = i

 e = j

 while (s < N && e >= 0) {

 print (A[s][e])

 s++

 e--

 }

print (new line)

}

}



Q2

You are given an integer matrix A, make all elements in a row and column as zero if $A[i][j] = 0$, Specifically, make entire i^{th} row and j^{th} col. zero.

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	8	0	4

(2,2)

row 1, col 3

make 1 row as zero
make 3 col as zero

	0	1	2	3
0	1	2	0	0
1	0	0	0	0
2	0	0	0	0

Note: all the nos are either zero or positive.

	0	1	2	3
0	1	2	3-1	4-1
1	5-1	6-1	7-1	0
2	9-1	8-1	0-1	4-1

1-1	12	13-1	14	15 ⁻¹	
2-1	10-1	0	16-1	17 ⁻¹	← flg=1
3-1	9-1	0	8-1	0	flg=1
0	4-1	5-1	6-1	7-1	flg=1
flg=1		flg=1			

0	12	0	14	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

TC: $O(MN + \underline{2MN} + 2MN)$

function makeZeroMat (A[M][N])

$O(\cancel{3MN})$
 $O(\underline{N \times M})$

for (i → 0 to N-1)

SC: $O(\underline{1})$

flag = 0

for (j → 0 to M-1) {

if (A[i][j] == 0)

flag = 1

}

if (flag == 1)

```

for (j = 0 to m-1) {
    if (A[i][j] != 0) {
        A[i][j] = -1
    }
}
}
}

```

```

for (j = 0 to m-1) {
    flag = 0
    for (i = 0 to n-1) {
        if (A[i][j] == 0) {
            flag = 1
        }
    }
    if (flag == 1) {
        for (i = 0 to n-1) {
            if (A[i][j] != 0) {

```

$$A[i][j] = -1$$

```
}  
}  
}
```

for ($i \rightarrow 0$ to $n-1$) {

for ($j \rightarrow 0$ to $m-1$) {

if ($A[i][j] == -1$) {

$A[i][j] = 0;$

```
}  
}  
}
```